

DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

APOL-651

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	651	R2FA	50A-APOL-ST3-E-7002-01	1 OF 2	0	CJ-M15W14-PR-A20-000503		
2	651	R2FA	50A-APOL-ST3-E-7002-02	2 OF 2	0	CJ-M15W14-PR-A20-000149		
3	651	2FA	50A-APOL-ST3-N-7002-01	1 OF 1	0			
4								
5								
6								
7								
8								
9								
10								
11								
12								

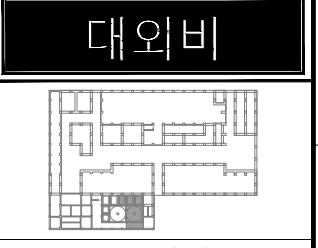
ITEM NO.	W14-T-4726
SERVICE	BRINE 2 nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.1 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 156.3 (EH 118.4)
HRT	min 34.1
MAT'L	P.C + FRP

ITEM NO.	W14-T-4727
SERVICE	BRINE 2 nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.2 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 160.1 (EH 121.3)
HRT	min 34.9
MAT'L	P.C + FRP

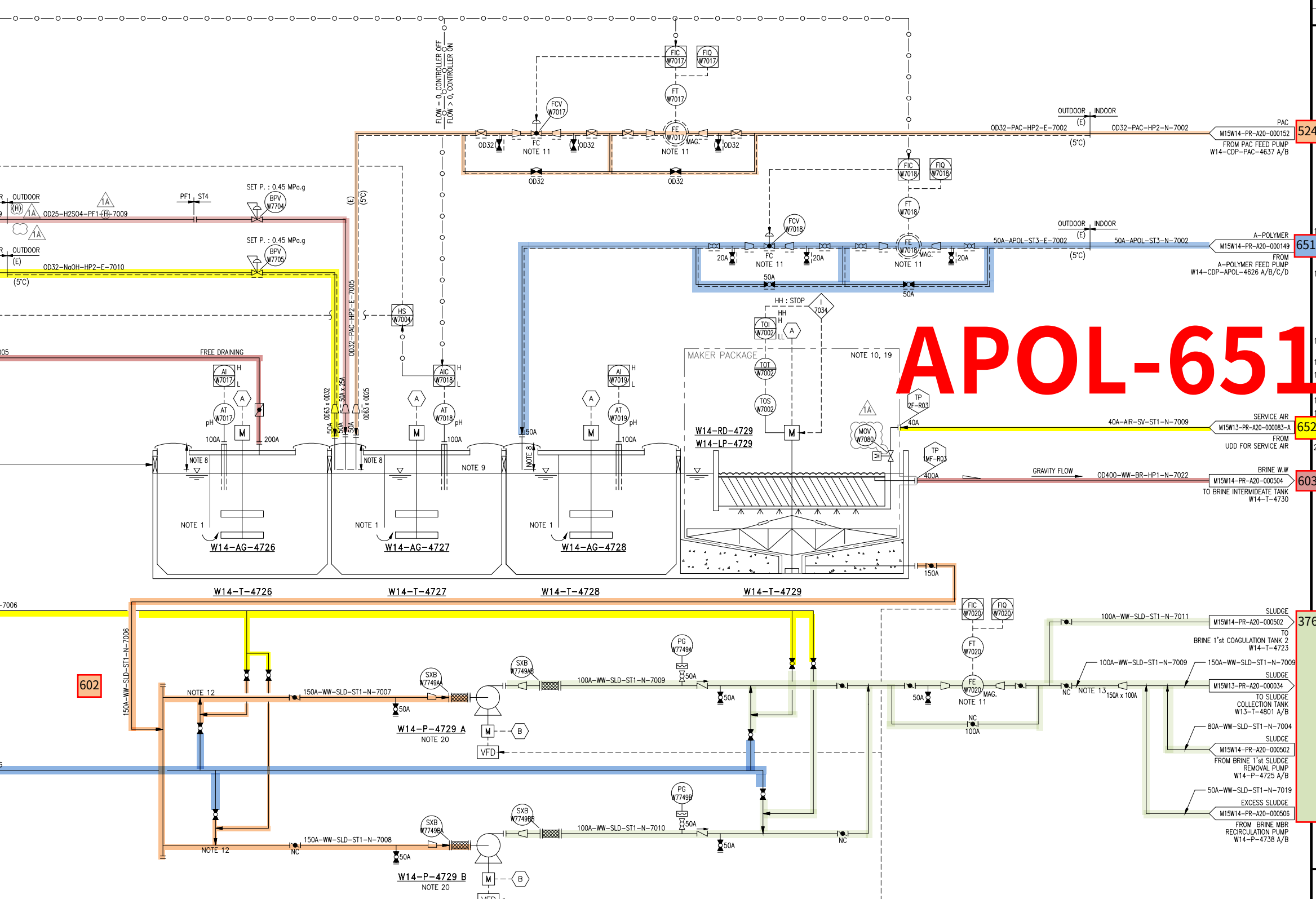
ITEM NO.	W14-T-4728
SERVICE	BRINE 2 nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 4.7 x 5.775 x 6.6 (EH 5.0)
CAPA.	m3 179.1 (EH 135.7)
HRT	min 39.1
MAT'L	P.C + FRP

ITEM NO.	W14-T-4729
SERVICE	BRINE 2 nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 9.225 x 9.225 x 6.6 (EH 5.0)
CAPA.	m3 561.7 (EH 425.5)
HRT	hr 2.0
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4729
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 5,000m3/d)
MAT'L	ABS+SS304



- NOTE**
1. BAFFLE 제형 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. MANHOLE : 2개 이상 설치한다.
MANHOLE COVER : FRP
 5. 온천 시 CLOSE 되는 배관은 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.
(SLAB으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 위에 갖추등의 상세사항은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정 예정.
 12. FLUSHING 용 WATER / AIR 배관은 45도 서면으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 또는 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.
 14. DELETED.
 15. 추후 VENDOR에서 결정한다.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. 상세한 장소 및 유지관리사항을 PACKAGE VENDOR 공급범위에 포함할 것.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.



ITEM NO.	W14-AG-4726
SERVICE	BRINE 2 nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (37)
MAT'L	STS316L/STS316L
POWER	kW (1)

ITEM NO.	W14-AG-4727
SERVICE	BRINE 2 nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kW (7.5)

ITEM NO.	W14-AG-4728
SERVICE	BRINE 2 nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kW (3.7)

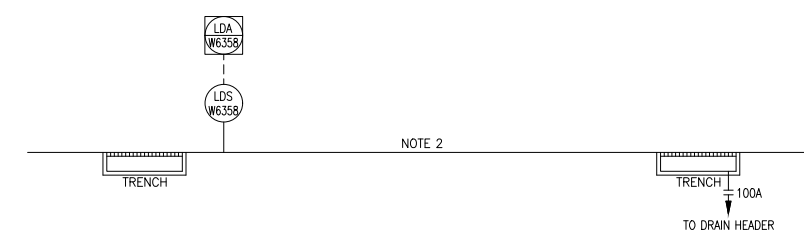
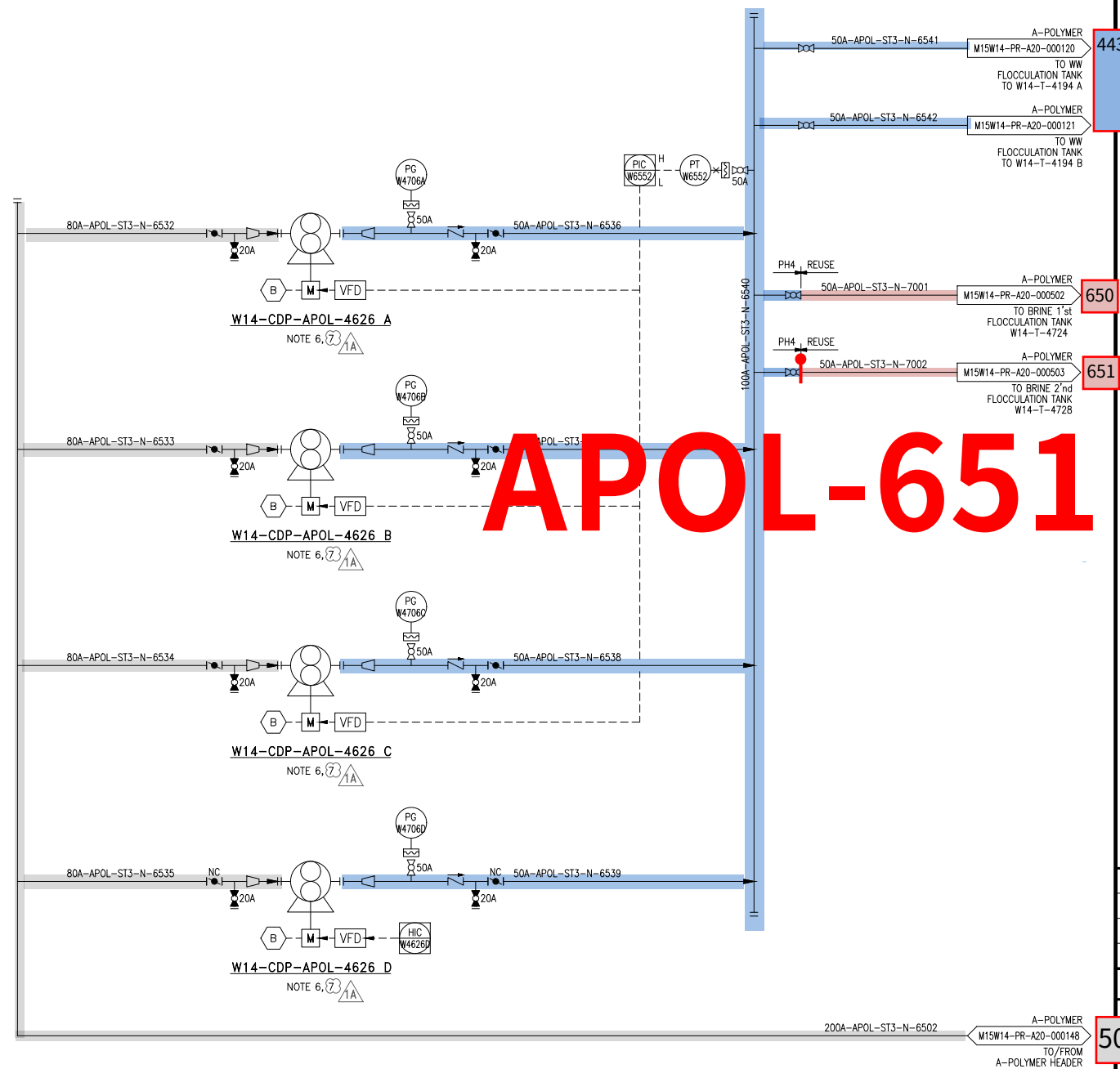
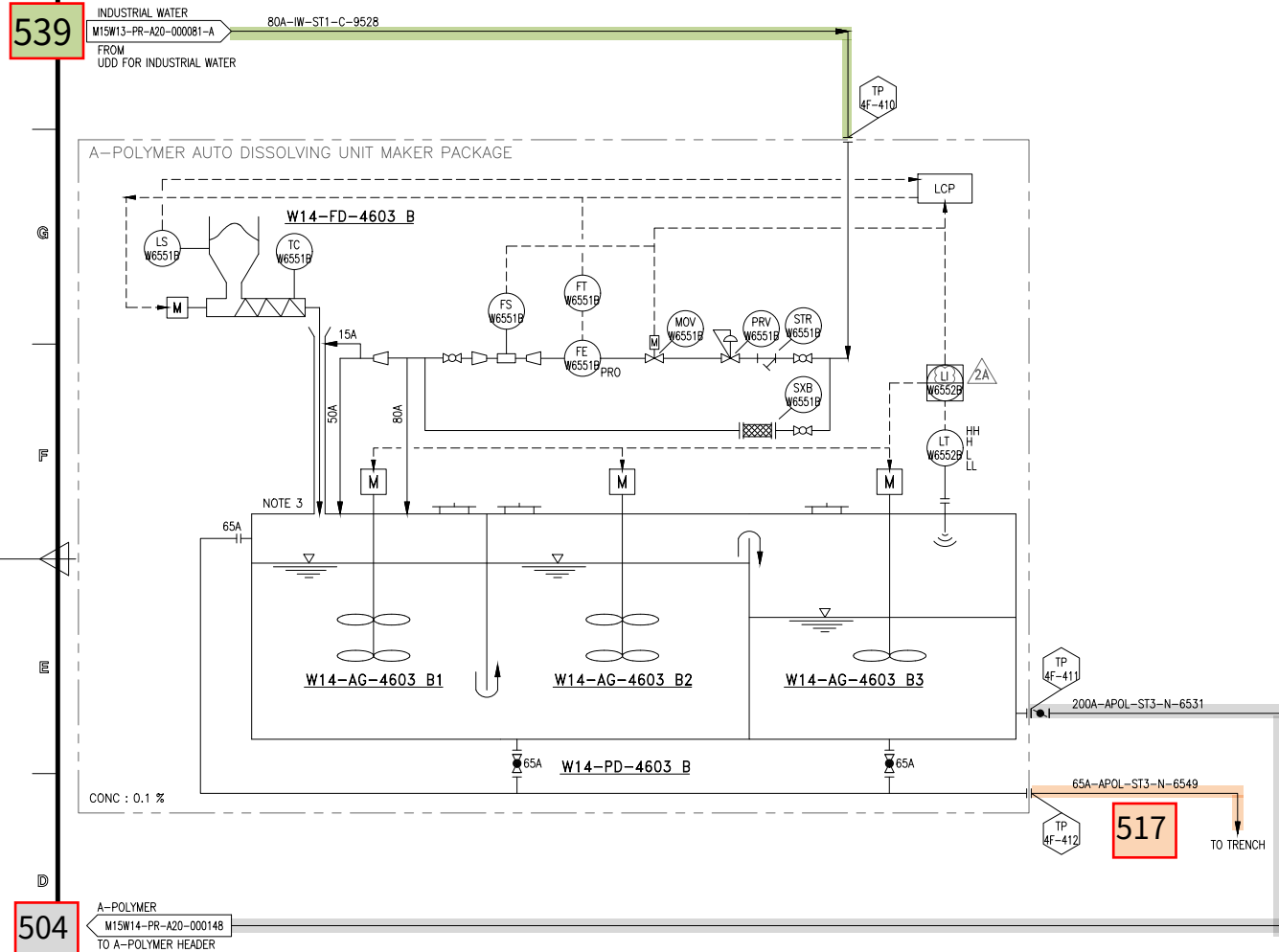
ITEM NO.	W14-RD-4729
SERVICE	BRINE 2 nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/d (0.079)
MAT'L	STS304/STS304
POWER	kW (1.5)

ITEM NO.	W14-P-4729 A/B
SERVICE	BRINE 2 nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (VVF), VORTEX
CAPA.	m3/hr.ea x mH 40 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kW (30)

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE
제	도	이 인 학
검	토	한 진 회
CHECKED BY		
승 인		
APPROVED BY		
제	작	사

SK 에코플랜트	
공 사 명	PROJECT TITLE
M15X Ph4 WWT Project	
도면명	DWG. TITLE
PIPING & INSTRUMENT DIAGRAM BRINE 2 nd COAGULATION & SETTLING TANK	
속	척
SCALE NONE	
날	짜
DATE 2025. 08. 14	
PROJECT NO.	22330D
도면번호	CJ-M15W14-PR-A20-000503
SHEET NO.	1A

ITEM NO.		W14-PD-4603 B	
SERVICE		A-POLYMER AUTO DISSOLVING UNIT	
Q'TY		1	
SIZE	W x L x H m	BY VENDOR	
CAPA.	kg/hr	20	
MAT'L		STS 304	
POWER	kW	7 } 2A	



A-POLYMER AUTO DISSOLVING UNIT MAKER PACKAGE : DESIGN BY SUPPLIER			
ITEM NO.	W14-FD-4603 B		
SERVICE	A-POLYMER FEEDER		
Q'TY	1		
TYPE	SCREW FEEDER		
CAPA. kg/hr	20		
MAT'L	STS304		
POWER kW	0.4		

ITEM NO.	W14-CDP-APOL-4626 A/B/C/D
SERVICE	A-POLYMER FEED PUMP
Q'TY	3 + 1
TYPE	ROTARY LOBE (WVF)
CAPA.	5 x 0.5
MAT'L	(NBR/GC250/SCM440)
POWER kW	3.7 (2A)

KEY PLAN (3F)

NOTE

1. 악품종류에 따라 LINE 색상을 구분하여야 한다.
2. POLYMER 용해장치 CURB 및 PUMP AREA에서 발생되는 DRAIN 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이동한다.
3. 상부 PLATFORM, HANDRAIL 설치 할 것.
4. AGITATOR MAT'L 표기 : SHAFT/BLADE
5. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
6. INTERNAL PSV 설치 할 것.
7. INTERNAL PSV SET PRESSURE : 0.65 Mpag

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수 정 REV.	설 명 DESCRIPTION	날 짜 DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 희
04 KED BY	
인	박 동 영
04 OVED BY	

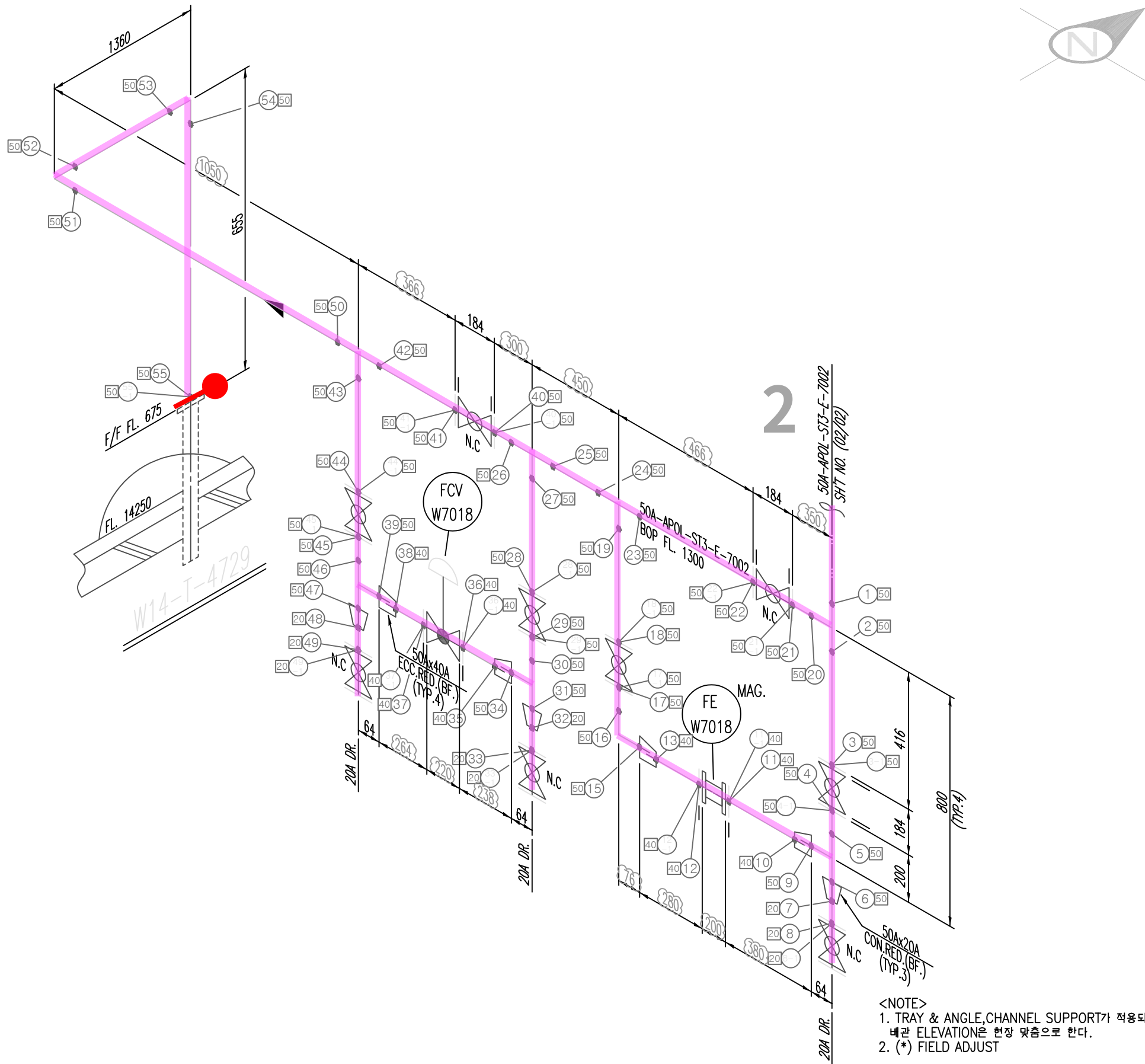
SK 에코플랜트

공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
A-POLYMER AUTO DISSOLVING
UNIT & FEED PUMP (2/2)

SCALE	NONE
DATE	2025. 08. 14

PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000149	2A



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	20A	0.3M	PIPE, STS 304TP, ERW, PE - SCH 10S
P	40A	1.3M	PIPE, STS 304TP, ERW, PE - SCH 10S
P	50A	7.5M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	3EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
T	50A	7EA	TEE EQUAL, STS304, ERW, BE - SCH 10S
RC	50Ax20A	3EA	REDUCER CON, STS304, ERW, BE - SCH 10S
RE	50Ax40A	4EA	REDUCER ECC, STS304, ERW, BE - SCH 10S
VB	20A	3EA	BALL VALVE, KS 10K, RF FLG'D, BODY-STSF304 / TRIM-STS304 / SEAT-PTFE, FB, LEVER
VB	50A	6EA	BALL VALVE, KS 10K, RF FLG'D, BODY-SSC13 / TRIM-STS304 / SEAT-PTFE, FB, LEVER
F	20A	3EA	FLG SO, KS 10K, RF, STS304
F	40A	4EA	FLG SO, KS 10K, RF, STS304
F	50A	13EA	FLG SO, KS 10K, RF, STS304
FB	20A	3EA	FLG BLIND, KS 10K, RF, STS304
G	20A	6EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	40A	4EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
G	50A	13EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	20A	6SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	40A	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	50A	13SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
LOTO	20A	3EA	LOTO
LOTO	50A	2EA	LOTO
INC	25tX20A	0.3M	관 보존재(고무발포) +알루미늄 CASING
INC	25tX40A	1.3M	관 보존재(고무발포) +알루미늄 CASING
INC	25tX50A	7.5M	관 보존재(고무발포) +알루미늄 CASING

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB110/VB2/VB210 * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VD10 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

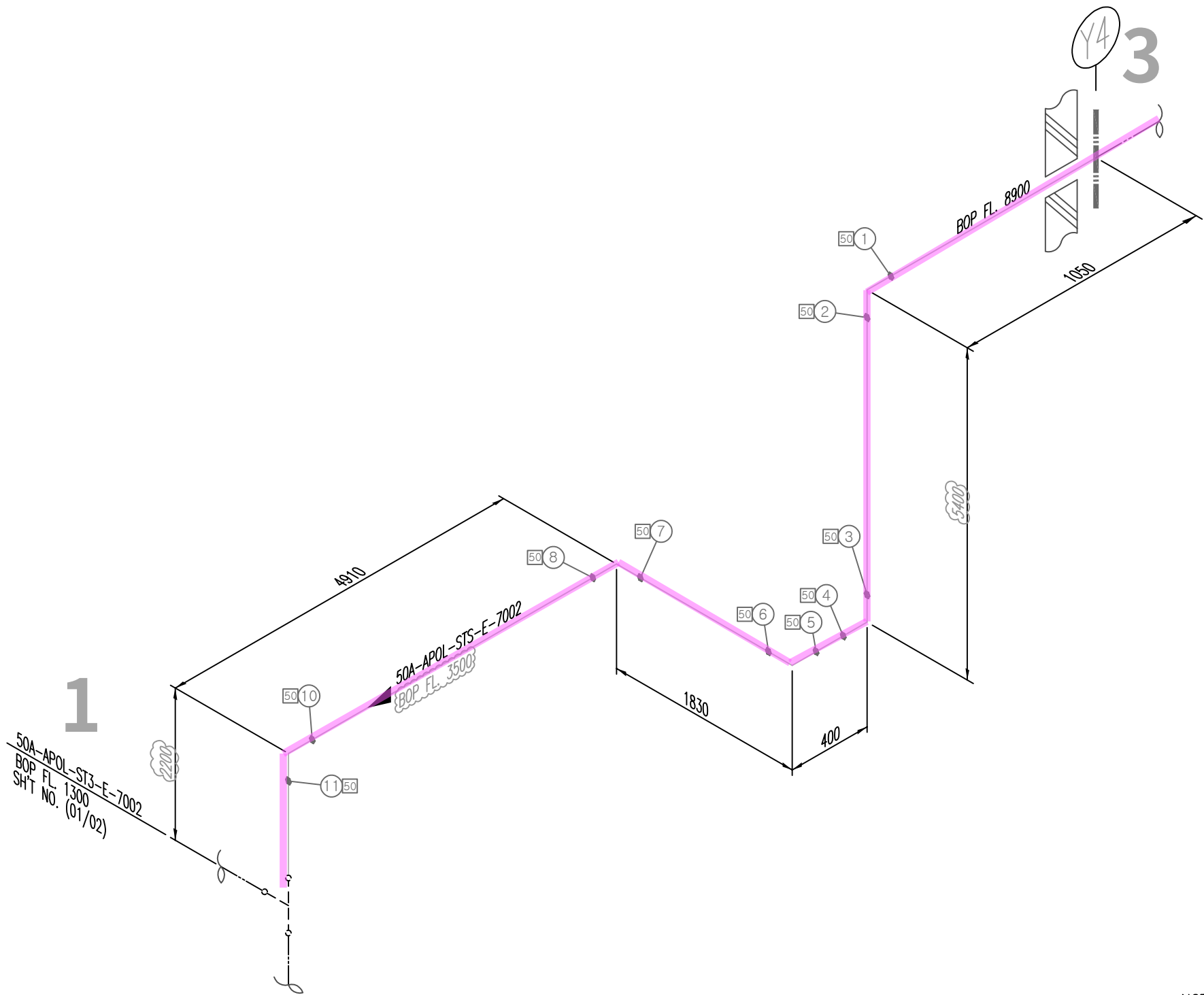
CLIENT	SK hynix	1
CONTRACTOR	SK ecoplant	PROJECT NAME M15X Ph4 Project
COMMODITY	APOL	JOB NO. 22330D
LINE NO.	50A-APOL-ST3-E-7002	SH'T NO. 01 / 02
		REV. 0

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14												
	THICK.	25	DESIGN	6.4	60	UNIT	R2FA												
			OPERATING	4	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-020016-A												
						REF. P. & ID	CJ-M15W14-PR-A20-000503												
						REV. NO.	25.01.24												
						DATE	APPROVED FOR CONSTRUCTION												
						DESCRIPTION	황래중												
						BY	안남후												
						CHK	박동영												
						APP'D													

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	50A	15.8M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	5EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
INC	25tX50A	15.8M	관 보온재(고무발포) + 알루미늄 CASING



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT

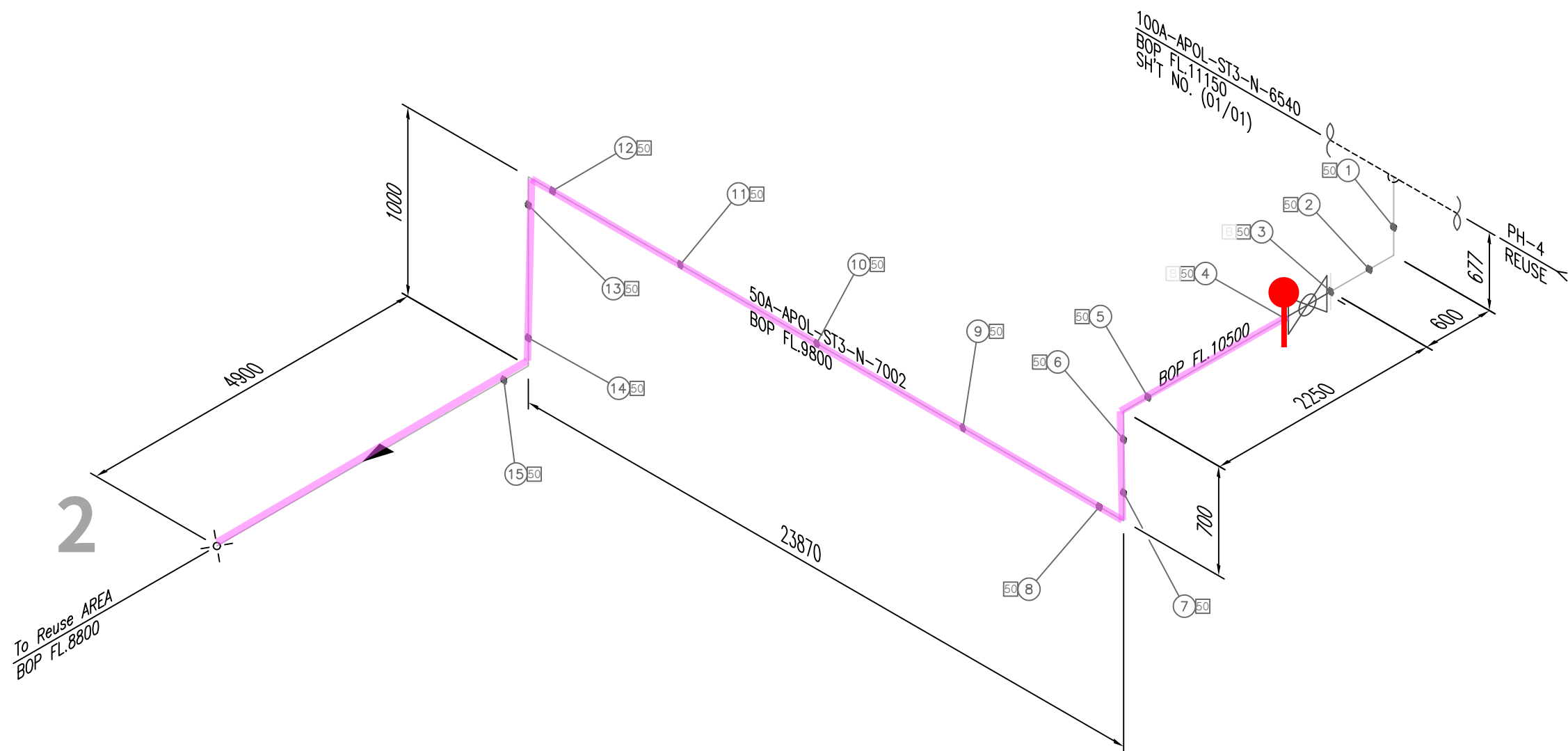


2



CONTRACTOR	PROJECT NAME
COMMODITY	JOB NO.
LINE NO.	SH'T NO.
	REV.

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14												
	THICK.	25	DESIGN	6.4	60	UNIT	R2FA												
			OPERATING	4	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-020016-A												
						REF. P. & ID	CJ-M15W14-PR-A20-000503												
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D								



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	5.7	60	UNIT	2FA										
			OPERATING	5.1	AMB	REF. PLAN DWG.		△	24.12.20	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			COMMODITY	JOB NO.
						CJ-M15W14-PI-A30-020012-H											
						REF. P. & ID		0	DATE	DESCRIPTION	BY	CHK	APP'D			LINE NO.	SH'T NO.
						CJ-M15W14-PR-A20-000149											
																50A-APOL-ST3-N-7002	01 / 01
																	REV.
																	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
ST3			
P	50A	34.0M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	50A	5EA	ELBOW 90 DEG LR, STS304, ERW, BE - SCH 10S
VB	50A	1EA	BALL VALVE, KS 10K, RF FLG'D, BODY-SSC13 /
			TRIM-ST304 / SEAT-PTFE, FB, LEVER
F	50A	2EA	FLG SO, KS 10K, RF, STS304
G	50A	2EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	50A	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS

CLIENT		3
		
CONTRACTOR		PROJECT NAME
		M15X Ph4 Project
COMMODITY		JOB NO.
APOL		22330D
LINE NO.		SH'T NO. REV.
50A-APOL-ST3-N-7002		01 / 01 0